
RAISE

Reasoning in Artificial Intelligence
and Software Engineering

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AI systems now write code, rank alerts and draft answers. **Very little of that output carries evidence that it is correct.**

A benchmark score summarises what happened last time. That is a useful thing to know. It is not a reason to trust the next answer.

RAISE builds what is missing: tests written to break a candidate patch, models that report their own error bars, and audits that say plainly where the evidence stops.

Three commitments behind every project

01

Evidence over confidence

A model sounding certain proves nothing. We make every claim answer to something outside the model: a test that tries to break it, a solver, a statistical check.

02

Measurement that means something

Before we report a number we ask what it measures and whether the data supports it. An evaluation is a measuring instrument. Instruments get calibrated.

03

Honest about limits

We build systems that can answer “inconclusive”. Knowing where the evidence runs out is more useful, and more honest, than a confident guess.

Three pillars, fourteen active directions

Software Engineering

5 DIRECTIONS

Agents across the lifecycle · AI governance ·
Static analysis and testing · Agent reliability ·
Quantum program testing

Artificial Intelligence and Machine Learning

5 DIRECTIONS

Evaluation and benchmarks · Uncertainty
and selective prediction · Efficient AI
systems · Adaptive vision · Retrieval and
multilingual NLP

Security

4 DIRECTIONS

Privacy in AI agents · Privacy-preserving
recommendation · Vulnerability detection ·
Trustworthy AI-generated code

A fourth strand runs alongside these: decision-making under uncertainty, where a forecast feeds a real optimisation step. ATM cash management and examination room allocation, both published this year.

Software Engineering

01 AI Agents in Software Development

Agents now touch requirements, code, tests and operations. We map what they actually do at each stage, then build evaluations that measure something real instead of a leaderboard delta.

02 AI Governance

Someone should be able to ask why a system accepted AI-written code. We record what justified the decision. A patch that passes the existing tests has proved little, so we write new ones built to break it.

03 Static Analysis and Testing

Nobody trusts an analyser that cries wolf. We rank alerts on evidence pulled from the repository and attach honest confidence to each, so the queue is worth working through.

04 Agent Reliability

An agent is a stochastic system, so we model it as one. Fitting execution traces to Markov chains yields error bars and a goodness-of-fit test. A single benchmark score yields neither.

05 Testing for Quantum Programs

You cannot debug a quantum program by running it and printing values. Correctness has to come from the type system and the logic, which is where our earlier work on separation logic pays off.

Artificial Intelligence and Machine Learning

01 AI Evaluation and Benchmarks

Using a model as a judge turns it into a measuring instrument, and instruments need calibrating. We score judges against verifiable ground truth and watch for benchmarks going stale through contamination.

02 Uncertainty and Selective Prediction

Some questions a system should decline. We build that refusal on finite-sample guarantees, set thresholds per group so fairness is not quietly traded away, and let error costs decide how cautious to be.

03 Efficient AI Systems

Five small models or one large one? Nobody can answer that without fixing the budget first. We run the comparison at equal cost, and ask how far aggregation goes before voting theory bites.

04 Adaptive Computer Vision

A frozen encoder meets a satellite sensor nobody trained it on. We adapt it at test time and without labels, reweighting spectral bands and routing adapters to the layers that need them.

05 Retrieval and Multilingual NLP

Legal text is full of amendments, citations and definitions that flat retrieval flattens away. We walk the knowledge graph alongside semantic search so that structure survives into the answer.

Security

01 Privacy in AI Agents

Persistent memory is the least audited part of an agent deployment. What one user's session writes, another user's session can read back. We built a benchmark and an audit suite so this gets measured, not argued about.

02 Privacy-Preserving Recommendation

A deployed recommender is asked for differential privacy, for calibration inside every group, and for the right to stay silent. Optimise these one at a time and each eats the others. We treat them as one constrained problem.

03 Vulnerability Detection

Every hypothesis is checked against what the repository actually shows, then marked Verified, Rejected or Inconclusive. We would rather report ten real bugs than a hundred maybes, so we tune for false discovery, not recall.

04 Trustworthy AI-Generated Code

Translated and repaired code has no author to ask. We put it through static analysis and generated tests, so accepting it rests on something firmer than the model's own say-so.

Four selected projects

TraceToChain

ISSRE 2026

Turns agent execution traces into an absorbing Markov chain, then refuses to report any number the statistics do not support. Tested on 2,459 real agent episodes.

GraphLedger

ASE 2026

Finds vulnerabilities across a whole repository and makes every finding show its work against real dependency and taint structure.

GraphMemShield

CIKM 2026

Measures what leaks between users in an agent's shared memory. Privacy stops being an argument and becomes a number.

LTM-RAG

ICDM 2026

Treats poisoning of retrieval-augmented generation as something that happens over time, not one query at a time. Attack success falls from 94.8% to 14.0%.

Ongoing research

The first wave of agent research asked what these systems can do. The questions we find interesting now are second-order: **not whether an agent can act, but when it should, and on whose evidence.**

When should it search?

Retrieval treated as a decision with a cost. Look something up only while the next lookup is worth more than it costs, and ask the user when asking beats guessing.

Whom should it trust?

Ten sources that copied one another are not ten pieces of evidence. We weigh evidence by independence, and stop untrusted content from ever carrying authority.

What should it remember?

Facts expire. We work on memory that tracks how long something stays true, and notices when fresh evidence contradicts what it stored.

How does it survive change?

Interfaces move, distributions shift, benchmarks go stale. A system should notice when it is running outside the conditions it was validated on.

Can it show its work?

Provenance you could hand to an auditor, tracing an output back through the data and the decisions that produced it. Sometimes with cryptographic guarantees.

Each question has work in preparation or under submission across search, security, data mining and machine learning venues.

A new laboratory built on established methods

28

PUBLICATIONS

19

PAPERS IN 2026

14

RESEARCH DIRECTIONS

RECENT — SOFTWARE ENGINEERING, DATA MINING, NLP, SECURITY

ICDM 2026 ×2 · EMNLP 2026 · CIKM 2026 · ASE 2026 ×2 · ISSRE 2026 · ACIVS 2026 ×3 · ACIIDS 2026 ×2 · IEA/AIE 2026 ×4

FOUNDATIONS — PROGRAM VERIFICATION, LOGIC, COMPLEXITY

NeurIPS 2024 · POPL 2022 · ICECCS 2020 · ICALP 2019 · APLAS 2018 · ESOP 2018 · ICFEM 2017 · FSTTCS 2016

Supervision and members

Dr. Xuan-Bach Le

HEAD OF LABORATORY • LECTURER, HCMUT

PhD, National University of Singapore, in program verification, under Aquinas Hobor and with mentorship from Anthony W. Lin.

Postdoctoral researcher, NTU Singapore, with Prof. Luke Ong.

Double degree in Computer Science and Pure Mathematics, NUS.

CURRENT STUDENTS • SELECTED

Thanh-Hai Tran Visual anomaly detection and risk-aware recovery

Thi-Hong-Cuc Le Repository-level vulnerability detection and explainable AI

Hoang-Quoc-Bao Hua LLM agents for static analysis and assumption validation

Huu-Vu-Phuong Tran Poison-resilient retrieval-augmented generation

Duc-Thuan Mai Longitudinal trust mining for retrieval pipelines

Van-Truong-Thinh Nguyen Depth-aware vision transformers for air quality

Tien-Anh Nguyen Test-time adaptation of vision-language models

Minh-Hieu Le Examination room optimisation, greedy and exact methods

RECENT GRADUATES • SELECTED

Khoa Phan • Phong Chung • Huu-Thanh Phan • Huynh Duy Khang

COLLABORATORS

Phat T. Tran-Truong (HCMUT) • Son Ha Xuan (RMIT)

Careful work, honest about its limits, and worth publishing.

Prospective students

Master's and undergraduate projects across all three pillars. You will own a problem, publish it, and learn to defend a claim with evidence.

Academic collaborators

We are looking for real agent traces and real static-analysis alert queues. Our methods are only worth as much as the data they meet.

Industry partners

Reliability audits, leakage assessments and evaluation design for deployed AI systems. Measurement you can put in front of a regulator.

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